

SudSceptor

HYDRODYNAMIC SEPARATOR

SEHDS750 - SEHDS3000 | DATA SHEET



OVERVIEW

SudSceptor is a hydrodynamic separator of proven and tested performance, sold as SEHDS750 to SEHDS3000. Designed for use within SuDS treatment trains, it gives certainty of water quality treatment and protects downstream controlled waters from sediment-bound pollutants, enabling enhanced amenity and biodiversity downstream.

Units are designed in accordance with the NJDEP protocol, best advice (NJDEP covers TSS only) and the British Water Code of Practice. A SudSceptor can be used on-line or off-line, with or without an external bypass unit.

PERFORMANCE & HYDRAULICS

SudSceptor removes sediment-bound contaminants and is made in a range of sizes to suit site flows and conditions. It removes over 50% of fine TSS (0 to 200 microns, median size 75 microns) at design flows and up to 99% of coarse solids of 0.1 to 0.4 mm.

MITIGATION INDICES	TSS	METALS	HYDROCARBONS
NJDEP*	0.5	-	-
SudSceptor	0.5	0.40	0.40

* NJDEP protocol, with best advice in part as NJDEP covers TSS only; British Water Code of Practice. Extrapolated values follow the British Water How to Guide: Applying the CIRIA SuDS Manual (C753), Simple Index Approach (version 7, section 4.3, 2021). The USA NJDEP has no regulatory role or authority in the UK and Ireland.

DESIGN COMPLIANCE

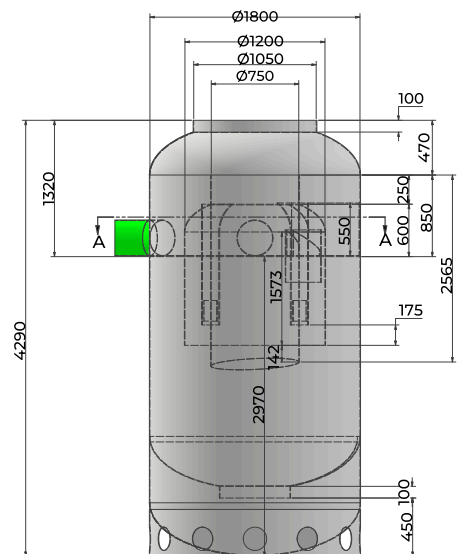
- NJDEP protocol 2015 and 2020
- DIBt
- British Water Code of Practice
- GRP models to BS 4994:1987, vessels and tanks in reinforced plastics

CHAMBER OPTIONS

SEHDS750 and SEHDS1000 units are made from BBA-certificated HDPE twinwall pipe. SEHDS1200, SEHDS1500, SEHDS1800, SEHDS2500 and SEHDS3000 are built in glass reinforced plastic (GRP). The range takes larger pipe connections, suiting sites with higher peak flows, and integrates easily with UK structured-wall drainage systems.

INSTALLATION

Installation design should consider traffic loading, ground conditions and variable groundwater levels. If in any doubt consult the scheme designer or engineer. A suitably qualified professional is required to sign off the installation.



SEHDS1800 elevation. Dimensions in mm.

NOT TO SCALE

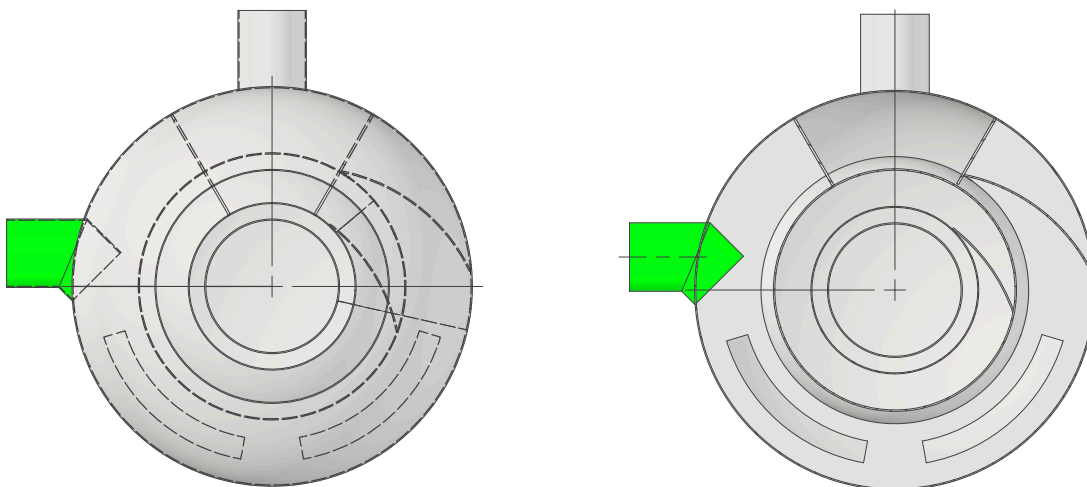
FEATURES & BENEFITS

- Single-piece unit
- Cannot air lock
- Wide variety of connection options
- Inlet deflector plate promotes the cyclic process
- Floatable materials and liquids are retained
- Settled solids are prevented from remobilising
- Uniform flow path via distributed upward flow
- Weir ensures optimised flow paths
- Easy maintenance
- 10-year warranty

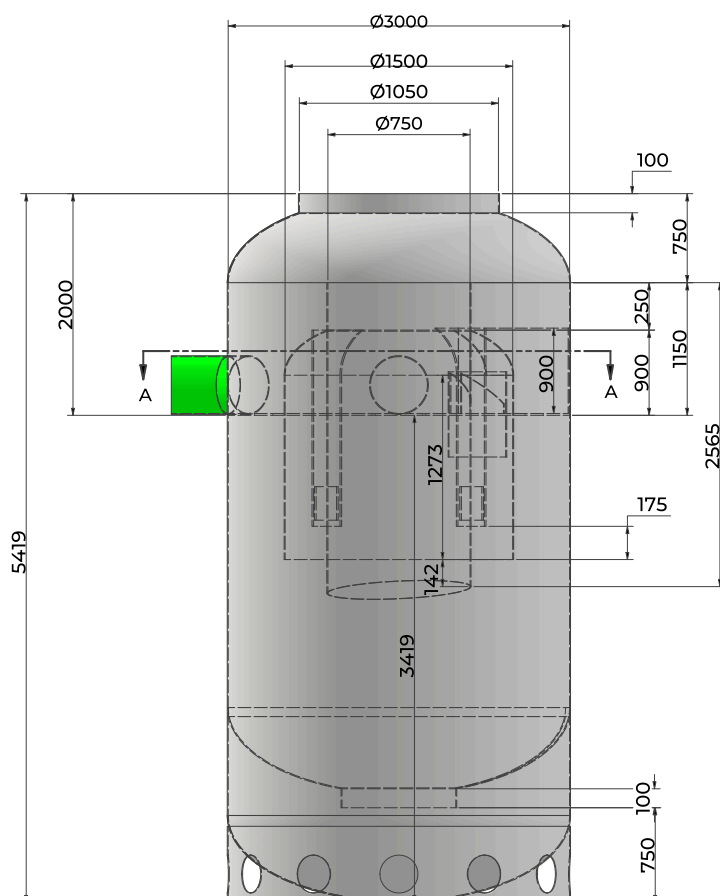
SPECIFICATION

Range	SEHDS750, 1200, 1800, 2500, 3000
Material	GRP from SEHDS1200; HDPE twinwall for SEHDS750
Treatment flow	32 to 259 L/s, by model
Connections	Up to Ø1200 mm inlet and outlet
Head loss	500 to 515 mm maximum
Filter	Optional RhinoPod polishing filter
Warranty	10 years

PLAN AND SECTION A-A PLAN | INLET SHOWN GREEN



ELEVATION



SEHDS3000

Chamber	Ø3000 mm GRP
Overall height	5419 mm
Inlet invert	2000 mm below top
Outlet invert	3419 mm above base
Max pipe	Ø1050 mm inlet and outlet
Sediment storage	4779 litres
Oil / debris storage	5024 litres
Treatment flow	189 L/s
Hydraulic capacity	655 L/s
Max head loss	500 mm

READING THE DRAWING

Drawing ADV-SEHDS3000 shows the Ø3000 GRP vessel with its Ø1500, Ø1050 and Ø750 internal tubes. The inlet invert is measured down from the top of the unit and the outlet invert up from the base.

NOT TO SCALE Reproduced from the SuDS Enviro drawing pack for illustration. Dimensions in millimetres. Confirm project dimensions against the order drawing.

1 INLET AND DEFLECTOR PLATE

Stormwater enters the inlet cylinder, where a deflector plate turns the flow and starts the cyclic process. The unit cannot air lock, and the inlet can be connected in a wide variety of positions.

2 SETTLEMENT IN THE SUMP

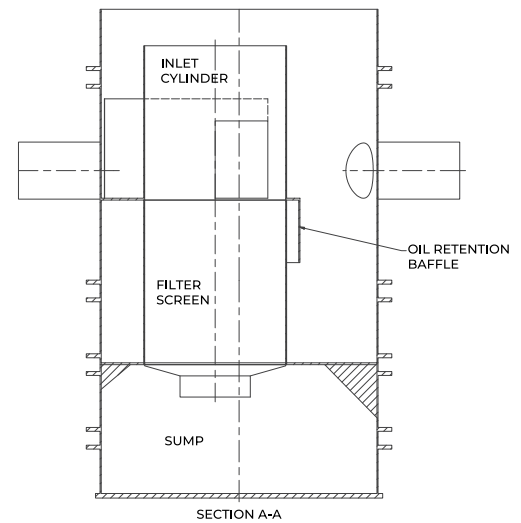
As the water circulates, sediment-bound pollutants settle out of the flow into the sump below. The settled solids are prevented from remobilising, so they stay in the sump until it is emptied during maintenance.

3 FLOATABLES AND OIL RETAINED

Floatable materials and liquids rise and are held back by the oil retention baffle, so they cannot pass to the outlet. The unit stores sediment and oil and debris separately; see the storage figures on page 4.

4 UPWARD FLOW AND WEIR

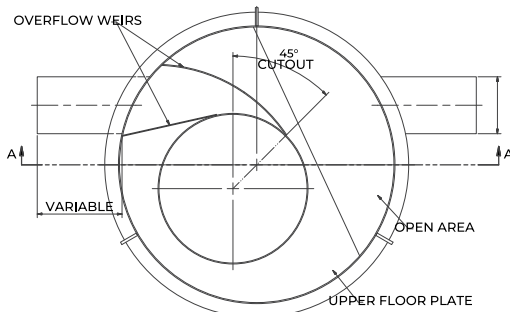
Treated water rises through the filter screen with a distributed upward flow, giving a uniform flow path, and leaves over the overflow weirs, which keep the flow paths optimised.



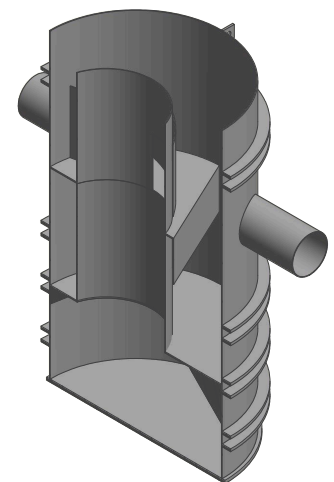
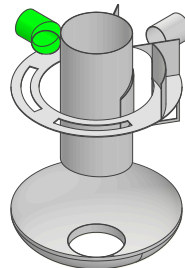
Section A-A. SudSceptor range drawing.

NOT TO SCALE

PLAN | WEIRS AND FLOOR PLATE



INTERNALS



Cutaway. SudSceptor range drawing.

TREATMENT TRAIN WITH RHINOPOD

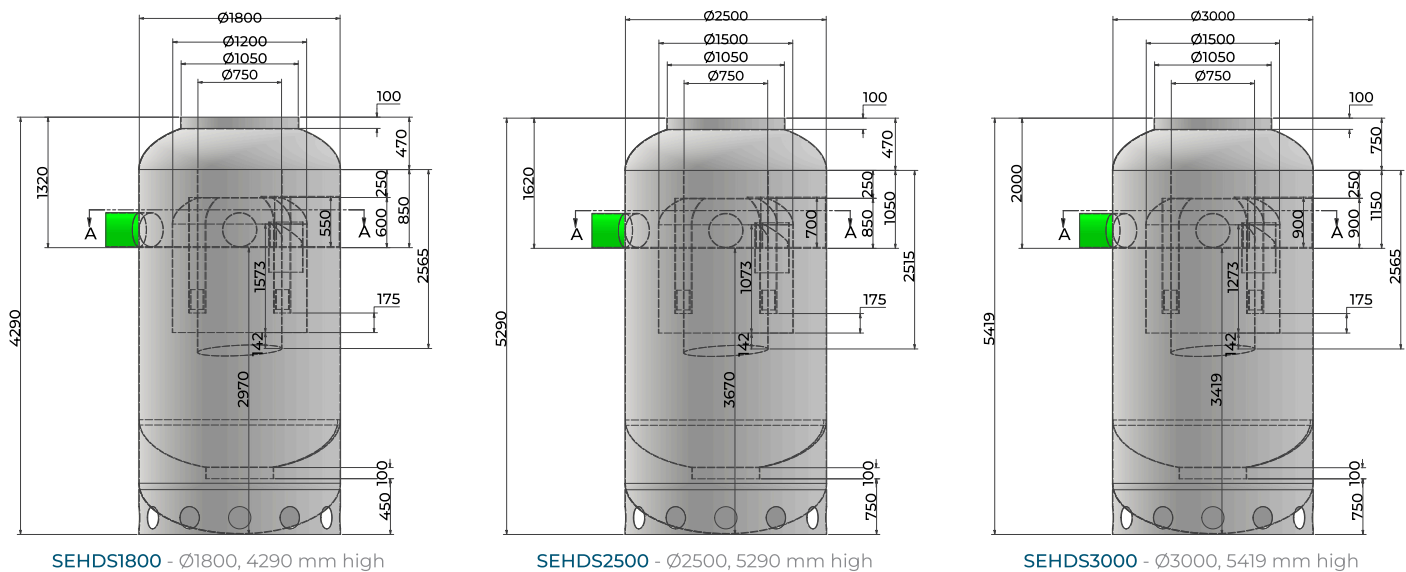
POLLUTANT	SUDSCEPTOR	RHINOPOD	COMBINED
Suspended solids	High	Low	Very high
Hydrocarbons	High	Medium	Very high
Zinc (Zn)	Moderate	High	High
Copper (Cu)	Low	High	High
Phosphate (PO ₄)	None	High	High
PAHs	None	Medium	High

Relative removal from independent lab and field evaluations in Sweden and across Europe, as published in the SudSceptor and RhinoPod brochure.

OPTIONAL RHINOPOD

The RhinoPod polishing filter captures a wide range of harmful substances, including oils, polycyclic aromatic hydrocarbons (PAHs), phosphorus, nitrogen and phosphate (PO₄).

It is capable of removing the 33 priority substances identified in the EU Water Framework Directive, as well as heavy metals such as cadmium, copper, zinc, nickel, lead and mercury.



SEHDS1800 - Ø1800, 4290 mm high

SEHDS2500 - Ø2500, 5290 mm high

SEHDS3000 - Ø3000, 5419 mm high

NOT TO SCALE Elevations are shown at different scales so each fits the column. The GRP models share one internal arrangement, scaled to each diameter.

CODE	CHAMBER	OVERALL HEIGHT	INLET INVERT FROM TOP	OUTLET INVERT FROM BASE	MAX PIPE	SEDIMENT STORAGE	OIL / DEBRIS STORAGE	TREATMENT FLOW	HYDRAULIC CAPACITY
SEHDS750	Ø750 HDPE	Performance and storage data on request							
SEHDS1200	Ø1200 GRP	2723 mm	TBC	1914 mm	Ø600	415 L	370 L	32 L/s	426 L/s
SEHDS1800	Ø1800 GRP	4290 mm	1320 mm	2970 mm	Ø600	1027 L	1462 L	73 L/s	222 L/s
SEHDS2500	Ø2500 GRP	5290 mm	1620 mm	3670 mm	Ø1200	3594 L	2332 L	259 L/s	426 L/s
SEHDS3000	Ø3000 GRP	5419 mm	2000 mm	3419 mm	Ø1050	4779 L	5024 L	189 L/s	655 L/s

Figures from the SEHDS1200, SEHDS1800, SEHDS2500 and SEHDS3000 data sheets. Overall height and inlet invert on the SEHDS2500 are standard and can vary. Maximum head loss is 500 mm (515 mm on the SEHDS2500).

INSTALLATION

- Bed and surround to suit the traffic loading and ground conditions
- Allow for variable groundwater levels and flotation
- Set inlet and outlet to the design inverts
- Use on-line or off-line, with or without an external bypass
- Have the installation signed off by a suitably qualified professional

COMPLIANCE

- NJDEP protocol 2015 and 2020
- DIBt
- British Water Code of Practice
- CIRIA C753 Simple Index Approach
- GRP to BS 4994:1987

SPECIFY SUDSCEPTOR

Send us the treatment flow, pipe sizes and invert levels and our team will select the model for your site.

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